Date:

GRAMMER: exp → exp addop term | term r3 ~~(Reduction 1 | r1)~~

addop → + | - r1/r2

term → term mulop factor | factor r5 ~~(Reduction 2 | r2)~~

mulop → * r4

factor → ( exp ) r6 | number r7 ~~(Reduction 3 | r3)~~

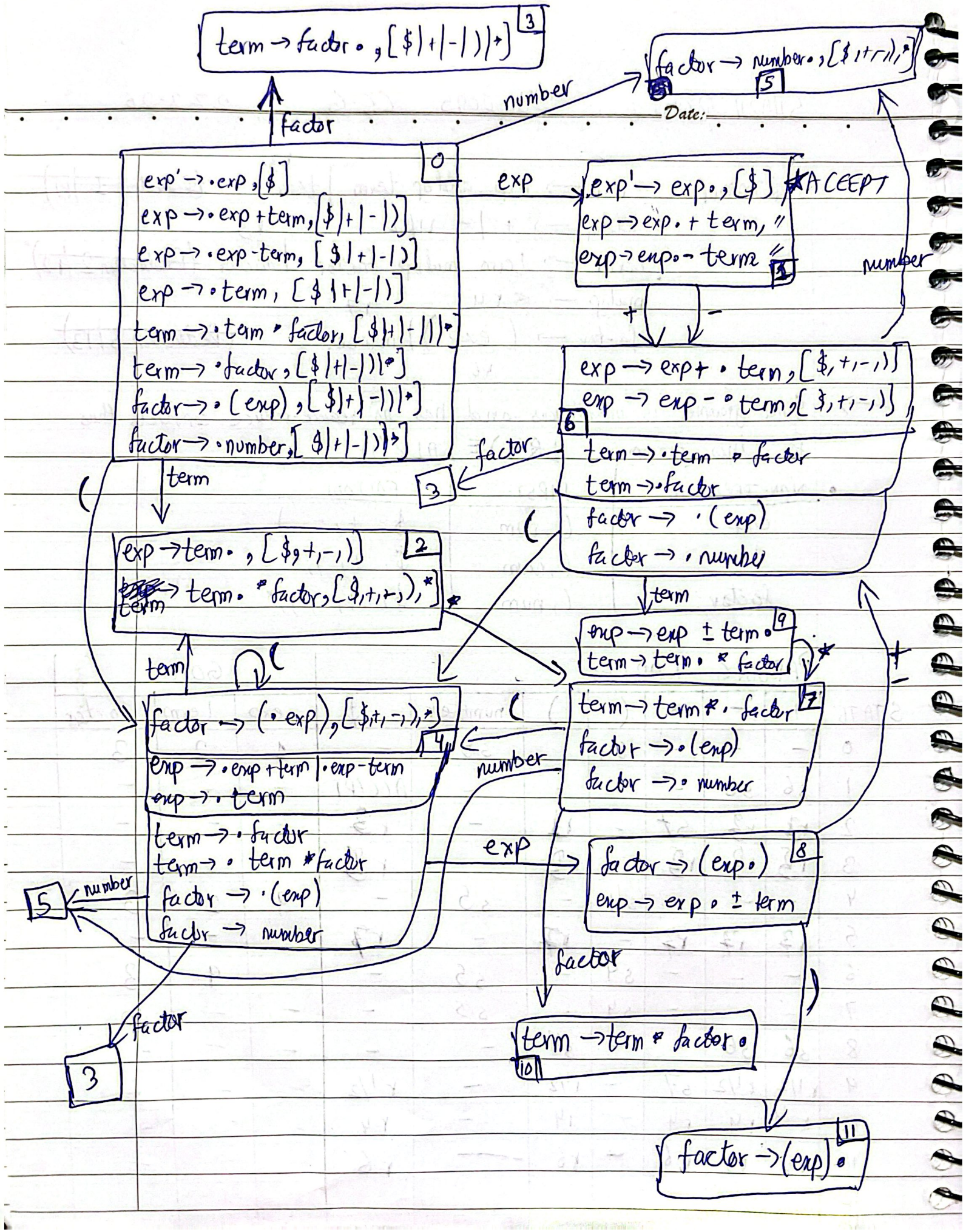
→ Since grammer is unambigious and has no reduce-reduce conflicts thus for this grammer LR(1) ≡ LALR(1)

| NON-TERMINAL | FIRST | FOLLOW |
|---|---|---|
| exp | (, num | $, +, -, ) |
| term | (, num | $, +, ), * |
| factor | (, num | $, +, -, ), * |

PARSING TABLE:

| STATE | + | - | * | ( | ) | number | $ | GOTO exp | term | factor |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | - | - | - | s4 | | s5 | - | 1 | 2 | 3 |
| 1 | s6 | s6 | - | - | - | - | ACCEPT | - | - | - |
| 2 | r3 | r3 | s7 | - | r3 | - | r3 | - | - | - |
| 3 | r5 | r5 | r5 | - | r5 | - | r5 | - | - | - |
| 4 | - | - | - | s4 | - | s5 | - | 9 | 2 | 3 |
| 5 | r7 | r7 | r7 | - | r7 | - | r7 | - | - | - |
| 6 | - | - | - | s4 | - | s5 | - | - | 9 | 3 |
| 7 | - | - | - | s4 | - | s5 | - | - | - | 10 |
| 8 | s6 | s6 | - | - | s11 | - | - | - | - | - |
| 9 | r1/2 | r1/2 | s7 | - | r1/2 | - | r1/2 | - | - | - |
| 10 | r4 | r4 | r4 | - | r4 | - | r4 | - | - | - |
| 11 | r6 | r6 | r6 | - | r6 | - | r6 | - | - | - |
| 12 | | | | | | | | - | | |
| 13 | | | | | | | | | | |

Scanned with CamScanner


term → factor •, [$|+|-|)|*] 3
factor
number
factor → number •, [$,+,-,),*] 5
Date:
0
exp' → • exp, [$]
exp → • exp + term, [$|+|-|)]
exp → • exp - term, [$|+|-|)]
exp → • term, [$|+|-|)]
term → • term * factor, [$|+|-|)|*]
term → • factor, [$|+|-|)|*]
factor → • (exp), [$|+|-|)|*]
factor → • number, [$|+|-|)|*]
exp
exp' → exp •, [$] ACCEPT
exp → exp • + term, "
exp → exp • - term "
1
+
-
number
exp → exp + • term, [$,+,-,)]
exp → exp - • term, [$,+,-,)]
6
term → • term * factor
term → • factor
factor → • (exp)
factor → • number
factor
3
term
exp → term •, [$,+,-,)] 2
term → term • * factor, [$,+,-,),*]
term
term
factor → (• exp), [$,+,-,),*] 4
exp → • exp + term | • exp - term
exp → • term
term → • factor
term → • term * factor
factor → • (exp)
factor → • number
number
5
factor
3
term
exp → exp ± term • 9
term → term • * factor
*
+
-
term → term * • factor 7
factor → • (exp)
factor → • number
number
exp
factor → (exp •) 8
exp → exp • ± term
factor
term → term * factor • 10
factor → (exp) • 11